

Devin Todd
Technical Portfolio

About me

Hi, I'm Devin. The lands and waters of the Pacific coast are my home. I share this home with many kin, so I often find myself working on stuff connected to sustainability.

When it comes to sustainability, I value action that has meaning and that can endure. That's led me to some proficiency at cultivating community across diversities of values, priorities, disciplines, and knowledges. Embracing coproduction processes is part of what helps me hack it – essentially this means empowering people.

The other part of my proficiency is having the transdisciplinary knowhow to facilitate coproduction. My foundation is more than a decade of engineering experience, including a Ph.D. and profession registration. I've grown from this foundation by knowing how to learn, and seeking the opportunities that value it – it's what keeps me engaged, and it's the thread that connects the works in this portfolio.

About this portfolio

This portfolio showcases some of initiatives that I have undertaken through the years. Some reflect short-term projects, while others reflect years of dedication.

To respect the intellectual property of project sponsors, descriptions have been limited to only general details.

FAQ

Why study engineering?

It seemed like a good idea at the time.

Why graduate studies?

It seemed like a good idea at the time, especially given a scholarship and the ~~once-in-a-lifetime~~ 2008 financial crisis.

Where is your Master's degree?

I skipped it because I am awesome.

Why study fuel cells?

They have merits, but it was not my first choice. A tsunami destroyed my original project's funding. No joke. I had to make do.

Why go into industry versus stay in academia?

I wanted to broaden my skills. The junior academic meat grinder is also very not-cool. I seek work that transcends these silos.

Why join a start-up after graduate school?

It is a great way to learn business-building without risking your own money.

Explain your role within New Leaf…

It was two-part: support internal projects; and provide the owner/investor technical consulting on other prospects.

What led you to join PICS?

I wanted to advance bigger initiatives further. PICS and the Solid Carbon team offered the opportunity.

Are you exploring new opportunities?

I always am. But I get proactive about it when the work is done or can run without my input. I also really dislike make-work.

What do you look for in new opportunities?

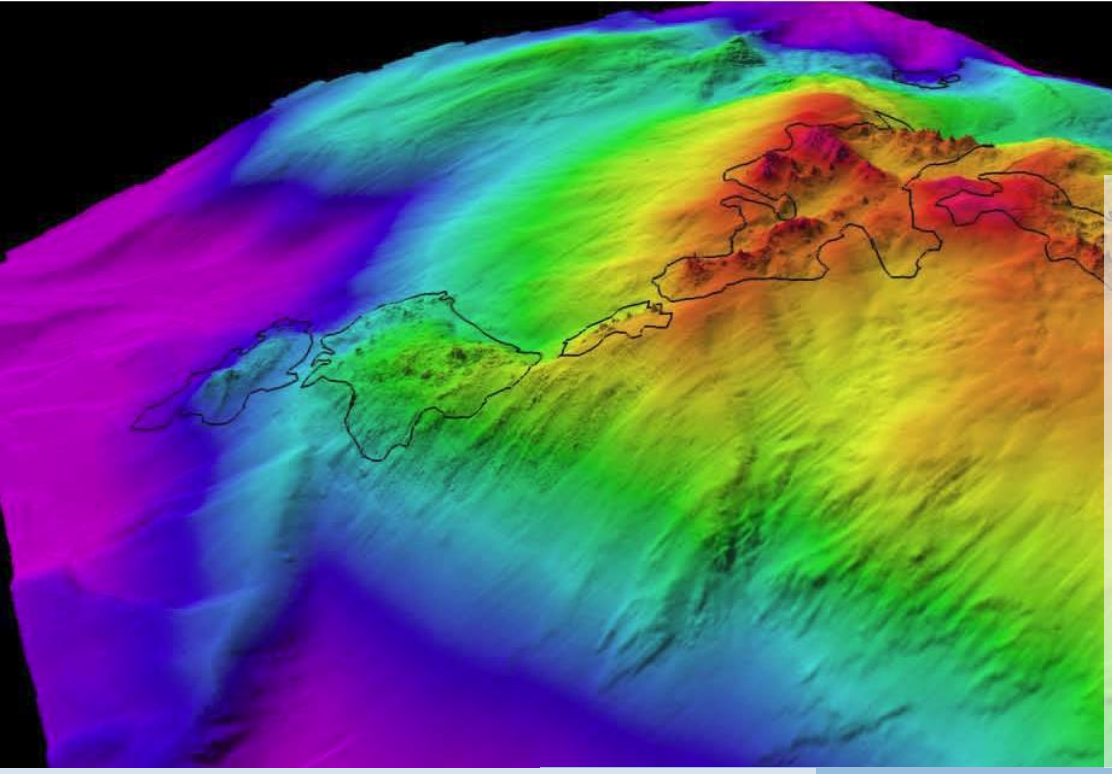
I will know it when I see it. Be prepared to demonstrate your values, integrity, and impact.

Why assemble a portfolio?

I needed a unique way to communicate the breadth of my experiences.

Tell me more about…

Email me at devin.g.z.todd@gmail.com to schedule a chat.



Mapping the Seafloor

Cellula Robotics Ltd.
Engineering Data Analyst
New Westminster, BC
Sept. 2007 – Dec. 2007

The client required bathymetry (3D reconstruction) of a subsea mining prospect at approximately 1200m depth and 0.2m grid resolution. This was to identify metal-rich chimneys and corroborate EM measurements.

A preliminary algorithm was expanded to fuse disparate sensor data files (e.g.: ship telemetry; ship-based ROV sonar positioning; ROV IMUs; multibeam sonar; and CTD). Code efficiency was improved and filter parameters were tuned to increase the speed and quality of output. Implementation was in MATLAB, with GIS for final rendering.

The mission's ROV telemetry and bathymetry were established, while the algorithm was developed to automate future analyses.

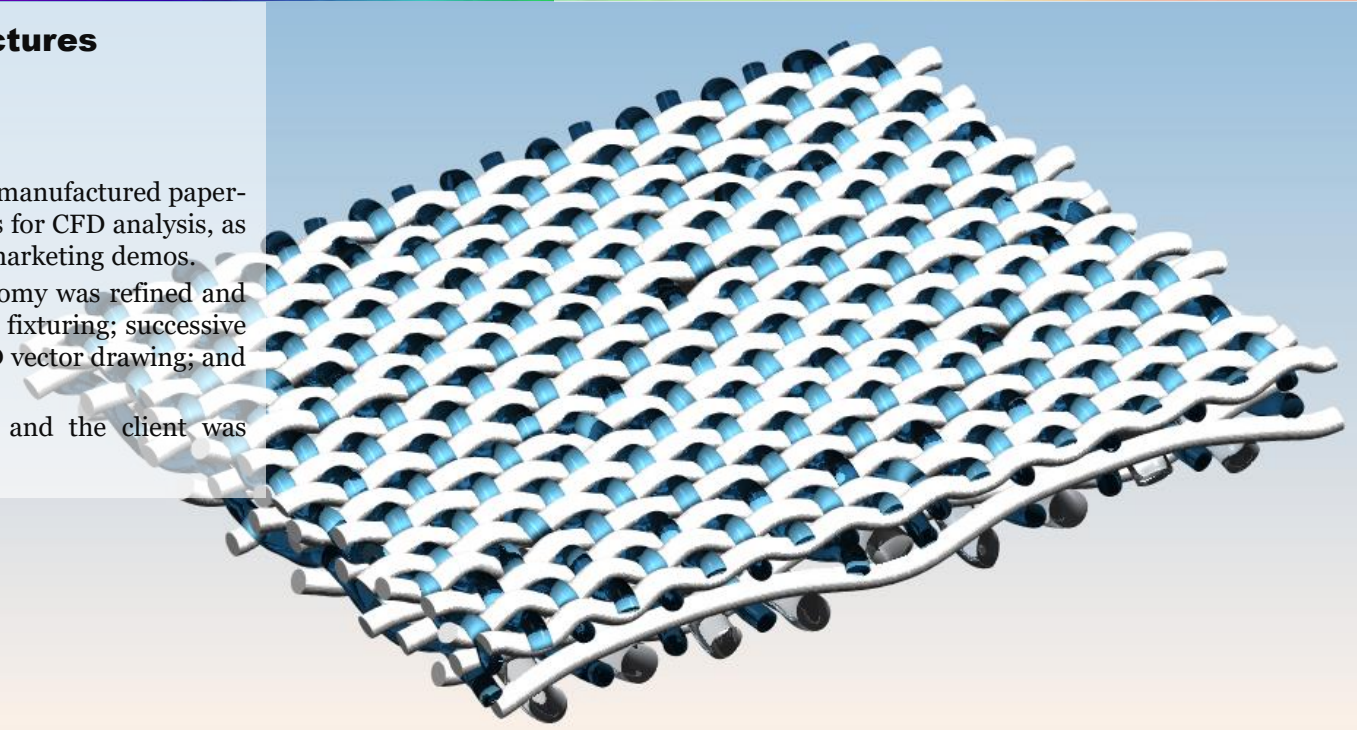
Reconstructing Complex Structures

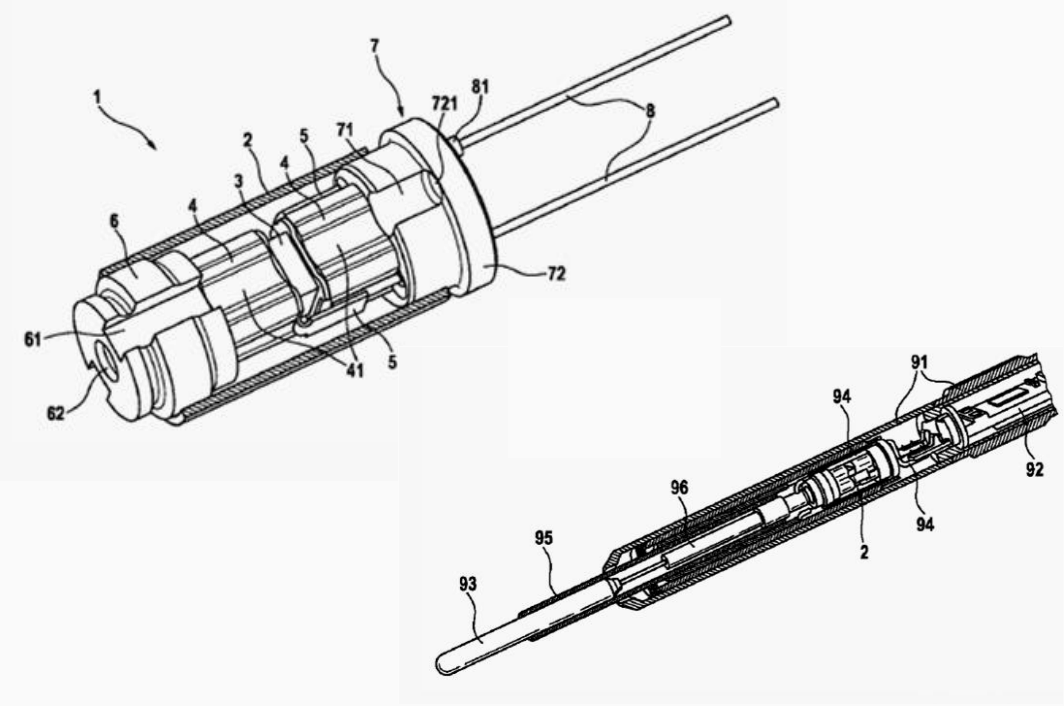
UBC Dept. of Mechanical Engineering
Undergraduate Research Assistant
Vancouver, BC
May 2007 – Aug. 2007

The client required 3D reconstructions of as-manufactured paper-forming fabrics. These were used by partners for CFD analysis, as well as 3D printed for PIV experiments and marketing demos.

The reconstruction method based on microtomy was refined and documented. The workflow included: sample fixturing; successive machining/slicing and imaging; AutoCAD 2D vector drawing; and SolidWorks 3D modelling.

Several fabric models were reconstructed, and the client was hosted for a workshop detailing the process.





Novel Automotive Sensors

Robert Bosch GmbH
CR/ARY3 Praktikant
Gerlingen, Germany
Jan. 2009 – Aug. 2009

Bosch required de-risking of a novel piezo sensor module for low-emissions automotive application in harsh (pressure and temperature) environments. De-risking comprised AST endurance, stress-to-failure, and sensitivity goals.

Experiments and corresponding test instruments were designed. Instruments featured piezo actuators, piezo sensors, charge amplifiers, and heating elements. NI hardware and LabVIEW were used for automated control and data logging. The workflow included: module preparation and testing; building of statistical models; and proposal of design refinements.

De-risking goals were successfully demonstrated and documented with six-sigma quality. The equivalent of a German engineering Bachelor's practicum was completed.

Hydrogen Infrastructure

Powertech Labs Inc.
Energy Systems Co-op Education Student
Surrey, BC
May 2008 – Aug. 2008

Clients required H2 technology systems, e.g.: fuel stations; storage and handling; embrittlement instruments; and electrolysis plants.

The role combined fabrication and coordination. I fabricated systems from P&IDs; managed part inventory; and communicated revisions between the fabrication and engineering shops. Work featured common metal fabrication processes, as well as specialized tube-fitting for gas pressures from 5,000-20,000 psi.

Several projects were built, including the HARP system (depicted), which uses surplus renewable electricity to produce H2 and supply a fuel station.



Build'n Race Cars

Formula SAE Team, UBC
Senior Member
Vancouver, BC
Nov. 2006 – Aug. 2011

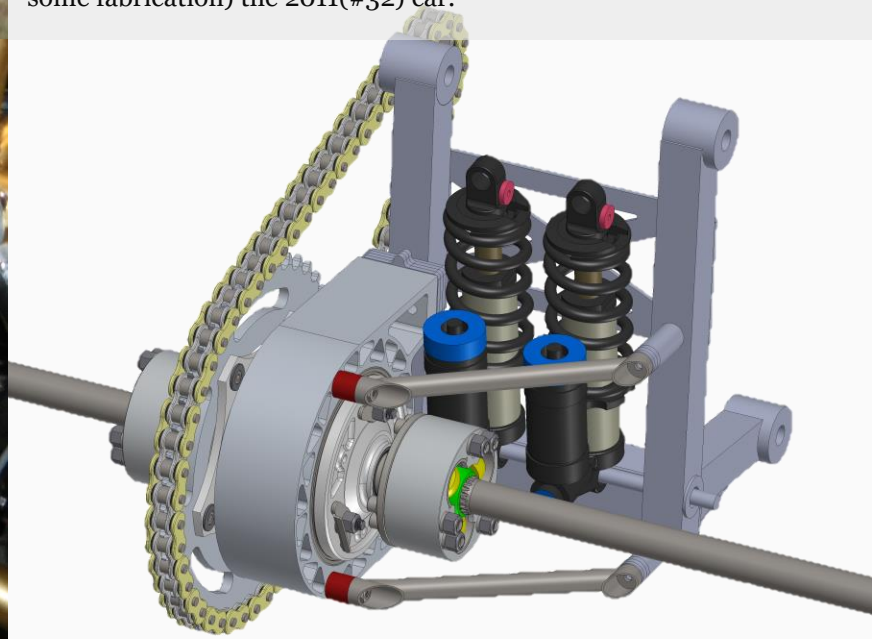
The objective was to build open-wheeled race cars for the Formula SAE student competition. Cars are evaluated on performance, design, and cost metrics. It is a design-build challenge with tight timelines and run outside the normal curriculum. Over the years my contributions evolved from a junior, to a leadership, to a mentor role.

My contributions focused on the drivetrain (although I lent a hand everywhere else). This encompassed everything following the output sprocket of a 600cc motorcycle engine: chain drive; limited-slip differential; drive shafts; CV joints; wheel hubs; and supports. Early designs featured VW parts which were modified for weight reduction. For the 2010 car, I had the mandate to redesign the whole system. The result was lighter, smaller, stiffer, and less leaky than the previous generation. The design was continued for 2011.

I developed design and fabrication skills across a spectrum of manufacturing technologies, e.g.: CAD/CAM/FEA/CFD; CNC cutting beds; metal fabrication; machining; composites work; etc. There was also design and contracting for specialty processes such a gear cutting and heat-treatment.

Outcomes comprise: fabricating the 2007(#16) car; designing and fabricating the 2008(#27) and 2010(#16) cars; and mentoring (with some fabrication) the 2011(#32) car.







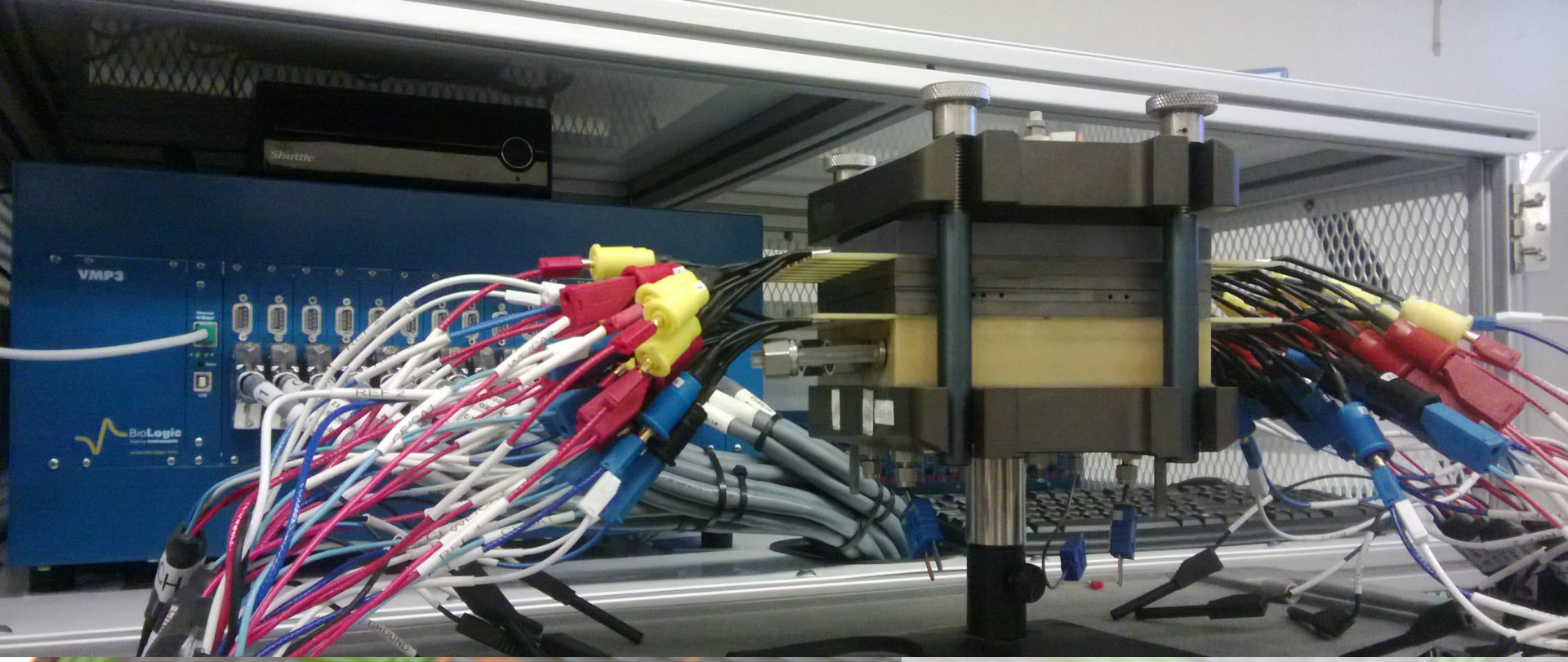
Novel Fuel Cell Materials

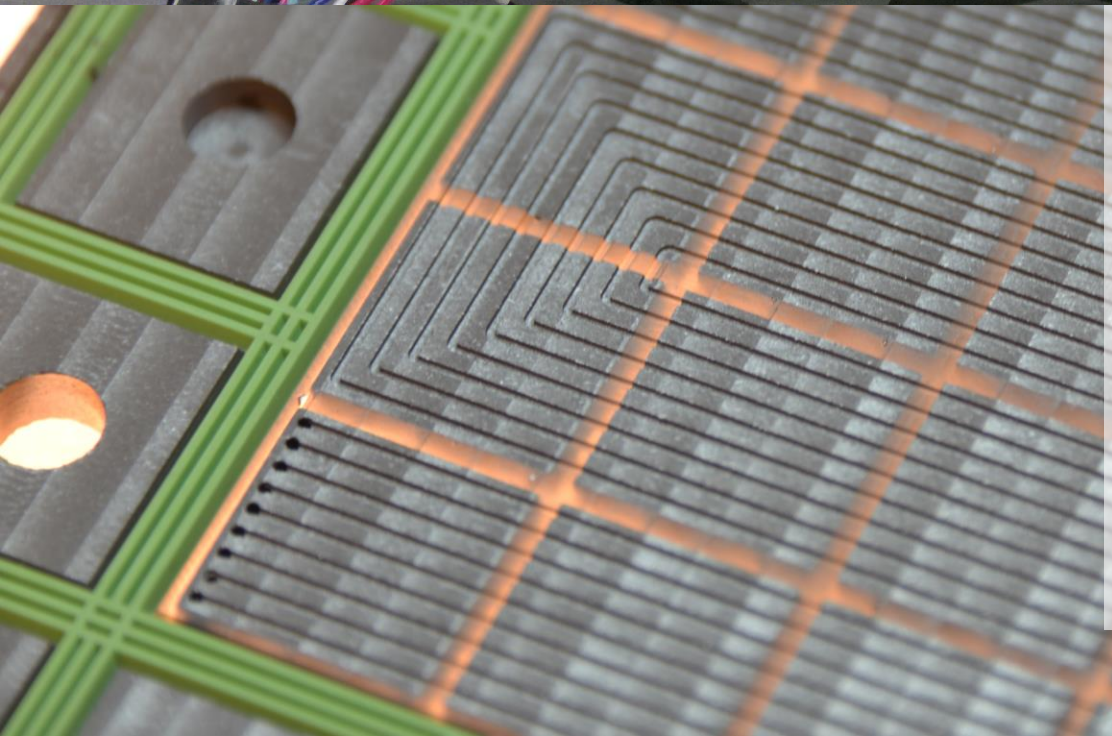
University of British Columbia
Ph.D. Candidate
Vancouver, BC
Sept. 2010 – Feb. 2016

The objective was to improve fuel cell performance via materials with engineered morphology. Contemporary transport layers (PTL/GDL) are based on repurposed carbon fiber paper and cloth technologies. Consequently, fuel cell designers are constrained to a small selection of products. A new approach was needed. I created novel hardware, methods and materials to improve performance through purposefully designed microstructures.

I adapted electrospinning technology to create layers with controlled morphology. The workflow comprised: polymer solution preparation; electrospinning; and thermal treatment. Layers were characterized ex-situ and in-situ.

My work spanned synthesis to application. It was the first demonstration of transport layers with controlled fiber diameters and alignments. It also revealed the importance of layer anisotropy for fuel cells in general, e.g. greater performance was observed with increasing alignment perpendicular to reactant distribution channels.





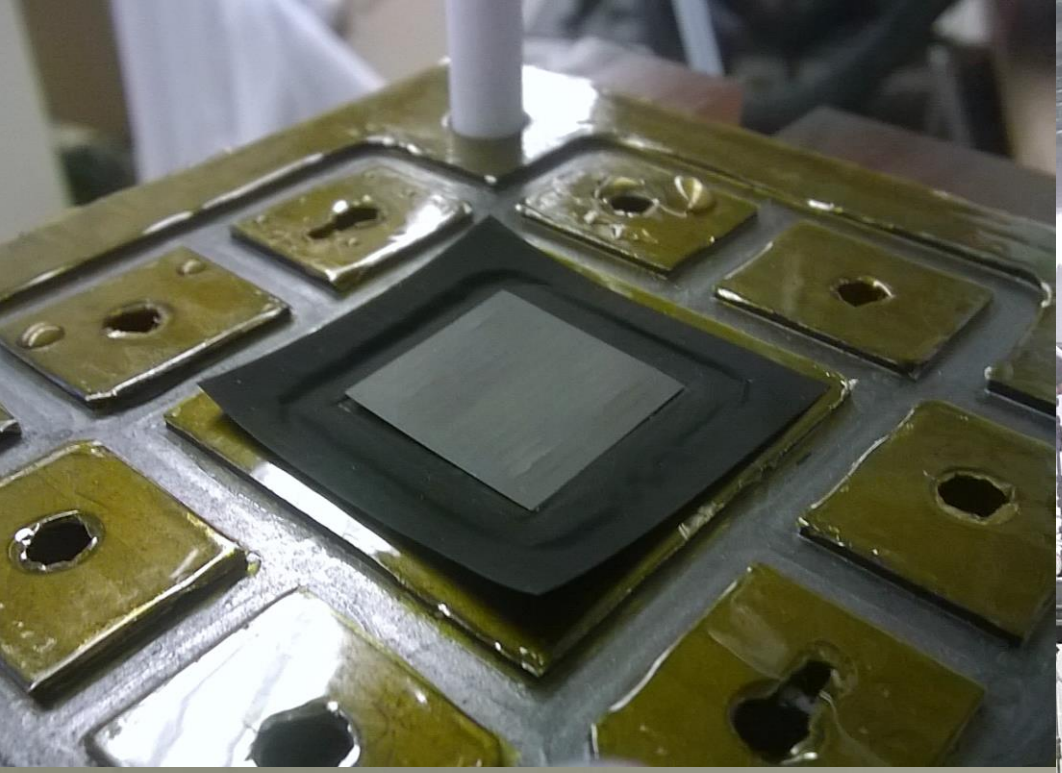
Segmented Cell System

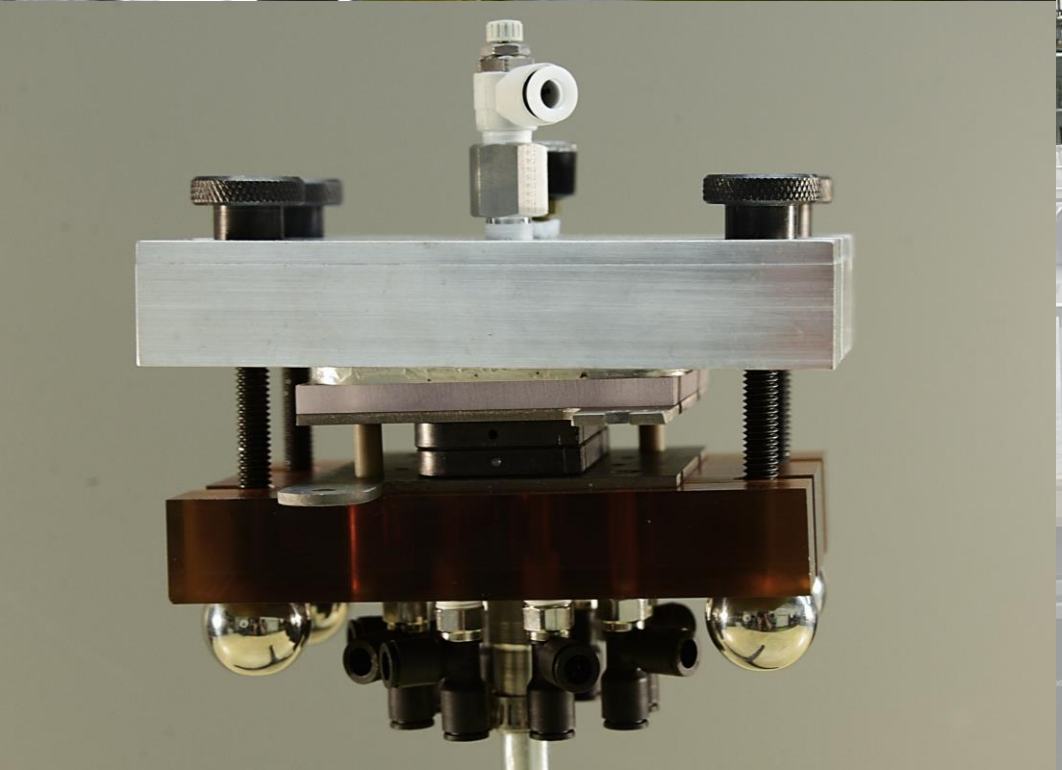
University of British Columbia
Ph.D. Candidate
Vancouver, BC
Sept. 2010 – Feb. 2016

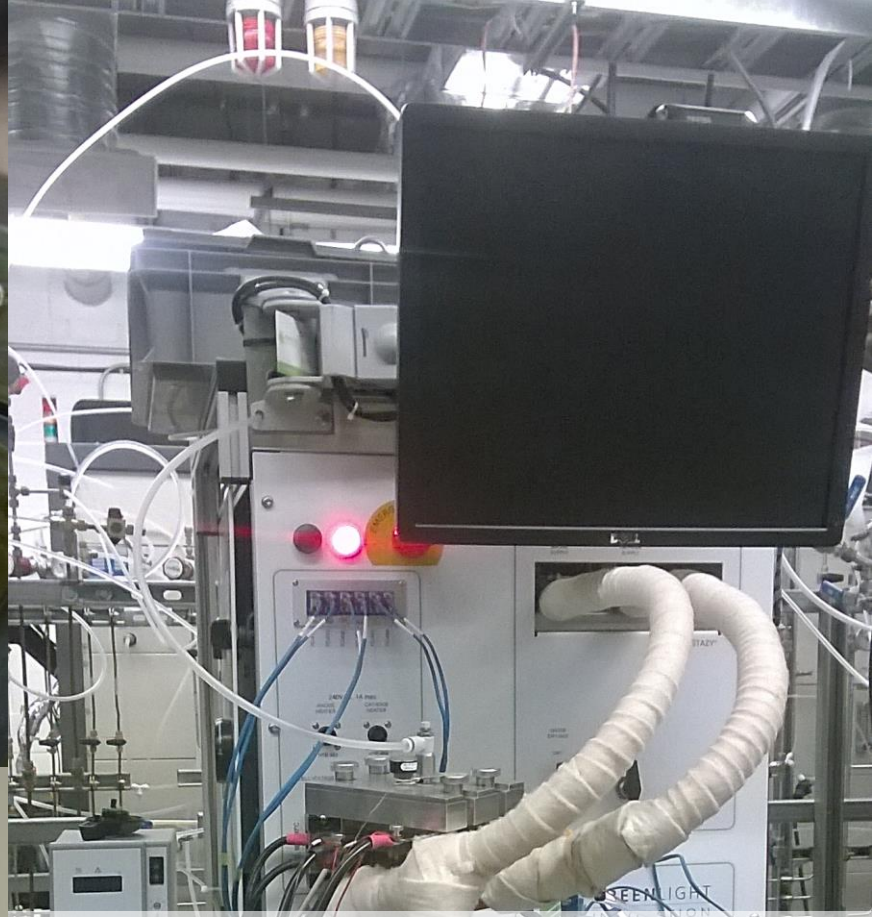
The objective was to measure spatially resolved fuel cell performance. In collaboration with M. Schwager, we developed the required hardware and demonstrated its application. My contribution focused on hardware development, whilst M. Schwager led the experimentation.

Fuel cell hardware was developed with electronically isolated segments, but common ionic connectivity (membrane). This included MEA segmentation via laser ablation. The cell was driven by a 16 channel potentiostat and current booster system with EIS capabilities. The overall system cost in excess of 100k$.

World leading capabilities were developed, comparable only those of our collaborators at Fraunhofer ISE. The first publication explored spatially resolved fuel cell catalyst layer degradation.







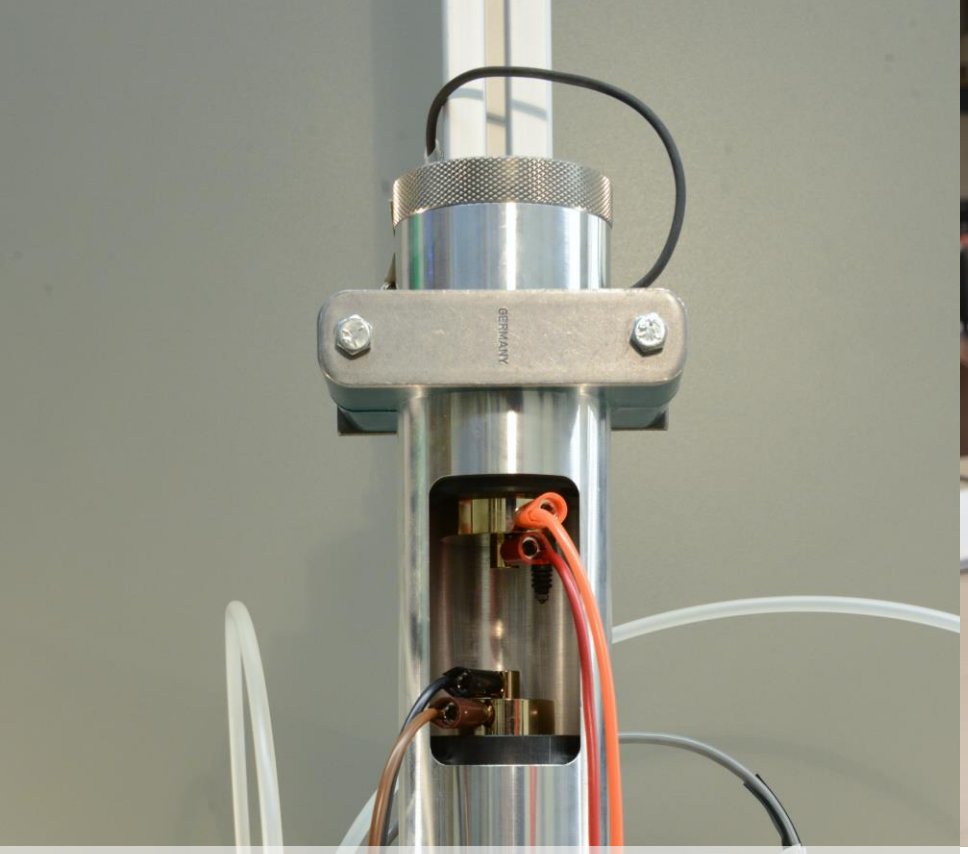
Small Area Fuel Cell

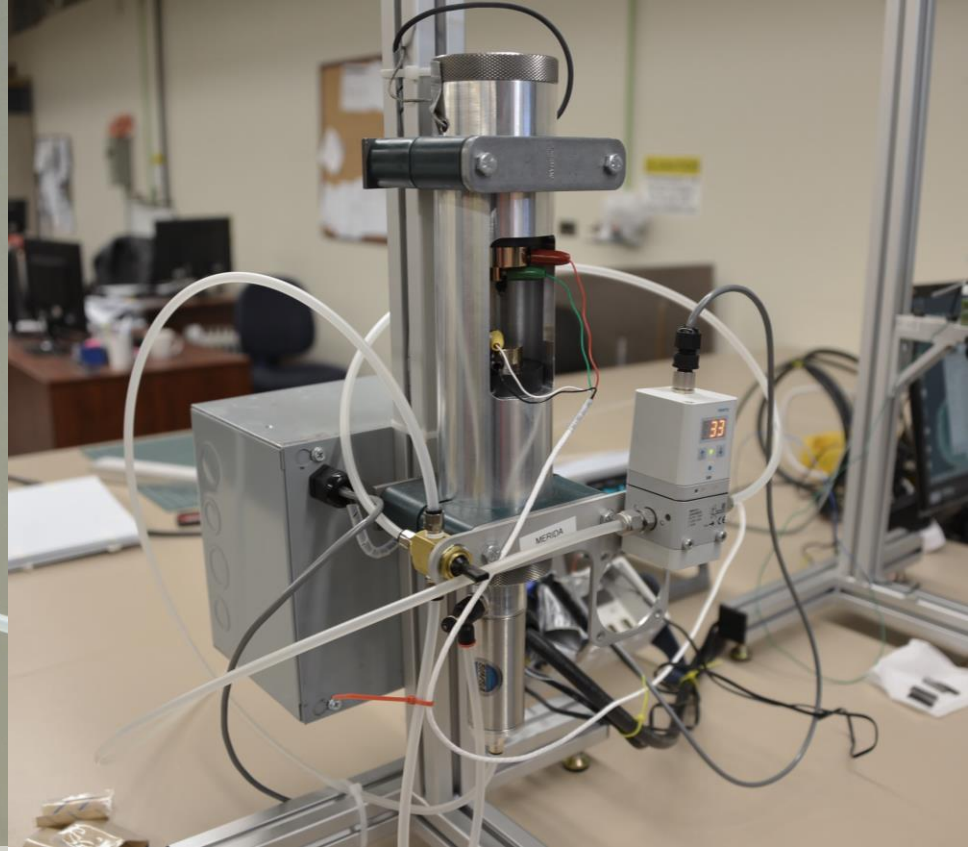
University of British Columbia
Ph.D. Candidate
Vancouver, BC
Sept. 2010 – Feb. 2016

The objective was to test the in-situ performance of the novel fuel cell materials that I created. This required a small active area test cell, as material size was constrained by manufacturing equipment. A review (and some testing) of commercial off the shelf test cells revealed that none were acceptable – so I developed my own.

A test cell was developed to accommodate cells from 2.25-5.00 cm^2. It featured liquid heating/cooling and pneumatic cell compression. It also featured several cost/time/performance improvements versus our larger cell design: drop-in cell assembly; reduced thermal mass; and fewer and simpler components.

Multiple cells were built to serve our group's research. Enabled by my other lab hardware investments, the design is fully in-house serviceable, manufacturable, and iterable.





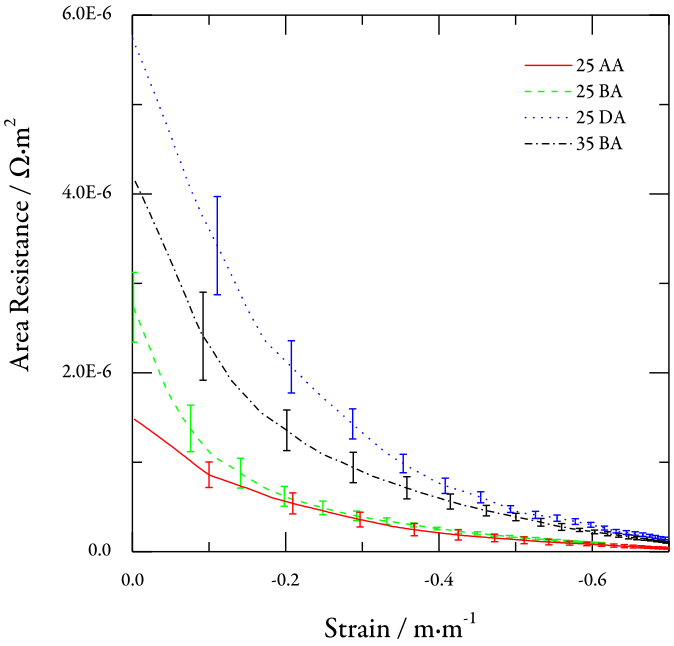
Electrical Resistivity (through-plane)

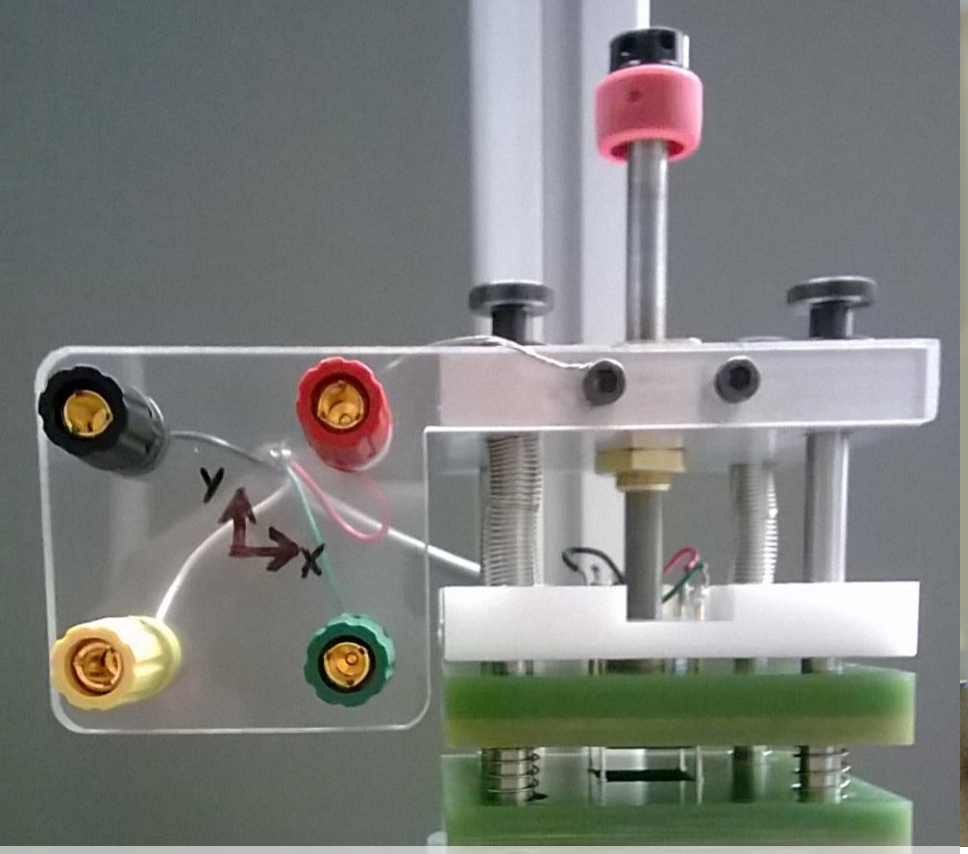
University of British Columbia
Ph.D. Candidate
Vancouver, BC
Sept. 2010 – Feb. 2016

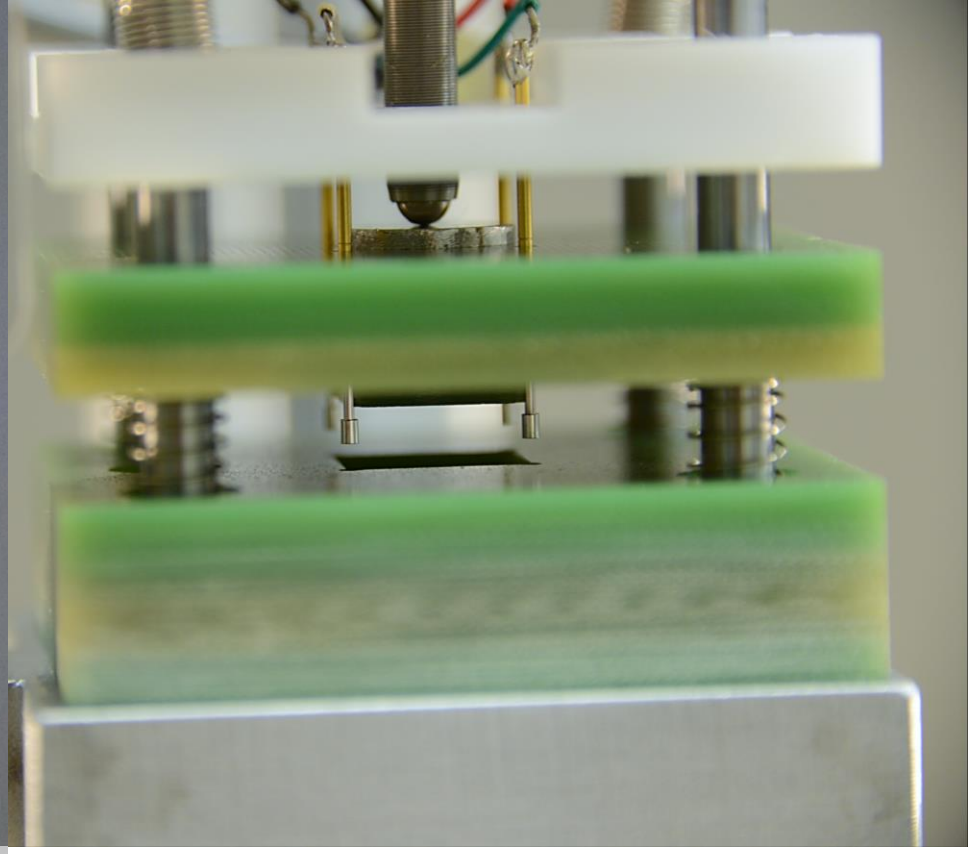
The objective was to measure the electrical resistivity of the novel fuel cell materials that I created. A review of the prior-art revealed that everyone was using inappropriate methods given the influence of material anisotropy. Consequently, I developed the right methods and hardware to measure transport layer resistivity as a function of mechanical strain.

The through-plane device featured gold-plated anvils which are pneumatically driven to engage a sample. Instrumentation comprised: force via a load cell; displacement via an LVDT; electrical resistance via high-resolution ohmmeter; and force control via a digital regulator (later addition). An NI DAQ and LabVIEW program were used to automate testing. The operator only needs to load/unload samples.

My novel approach was published in peer-reviewed literature. The system also provided quantitative evidence of my novel synthesis method's capacity to control morphology. World leading capabilities were developed, and were contracted to local fuel cell industry.







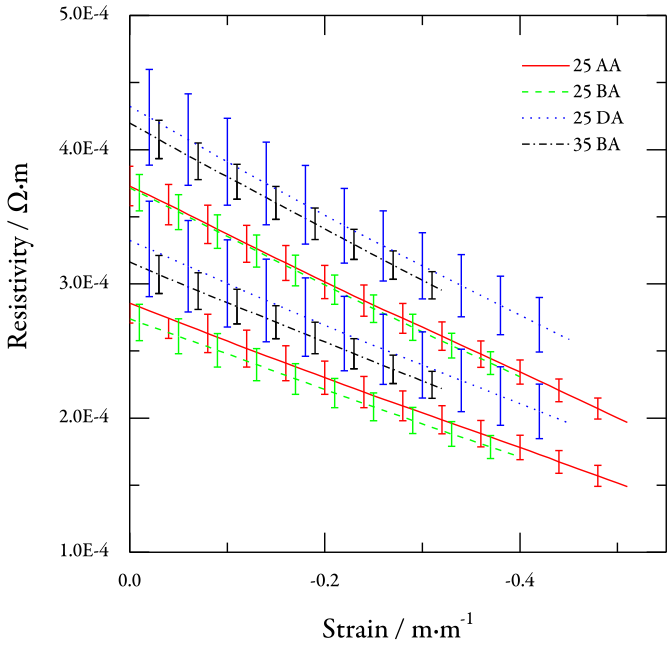
Electrical Resistivity (in-plane)

University of British Columbia
Ph.D. Candidate
Vancouver, BC
Sept. 2010 – Feb. 2016

The objective was to measure the electrical resistivity of the novel fuel cell materials that I created. A review of the prior-art revealed that everyone was using inappropriate methods given the influence of material anisotropy. Consequently, I developed the right methods and hardware to measure transport layer resistivity as a function of mechanical strain.

The in-plane device featured a square array of probes for resistivity sensing, and a screw-driven non-conductive anvil to apply mechanical strain (measured via a digital indicator). A high-resolution ohmmeter coupled to an NI DAQ running LabVIEW was used for logging resistivity. The operator loads/unloads samples, and is guided by the program through the semi-automated testing procedure.

My novel approach was published in peer-reviewed literature. The system also provided quantitative evidence of my novel synthesis method's capacity to control morphology. World leading capabilities were developed, and were contracted to local fuel cell industry.



(Re)Building and Managing a Lab

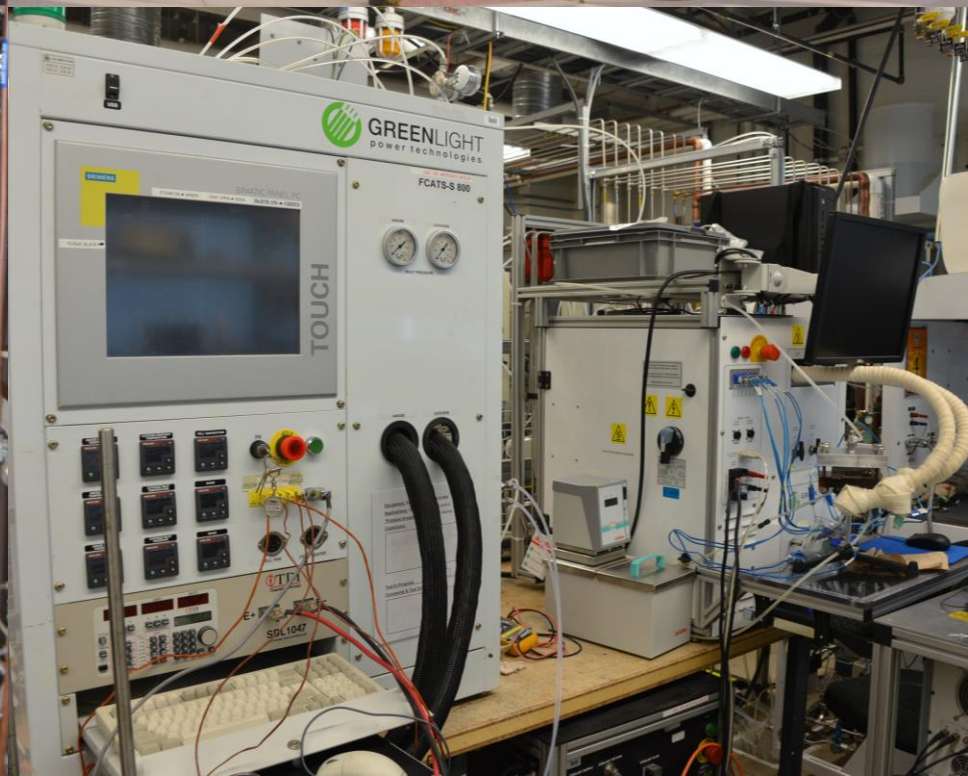
University of British Columbia
Ph.D. Candidate
Vancouver, BC
Sept. 2010 – Feb. 2016

I inherited a disused H2 technology laboratory following the PI's sabbatical. It needed work before it could serve my research needs, so I took on responsibility for rebuilding and managing it. Business cases were developed for new equipment and upgrades. Meanwhile, I led: the day-to-day of user safety and training; and the installation, troubleshooting, and maintenance of equipment. Overall, I oversaw several 100k$ of facilities, including:

- Built-in gas/fluid infrastructure (H2, N2, O2, air, DI water, etc.)
- Four fuel cell test stations
- Electrical load banks and impedance spectrometers
- Atmosphere-controlled furnaces
- CNC mill
- And more…







Tech Startup Hustle

New Leaf Management Ltd.
Team Member
Vancouver, BC
Apr. 2016 – Feb 2020

Following my Ph.D., I joined a startup to gain frontline experience in the business of R&D. New Leaf presented a unique opportunity to work two roles: as a member of the startup team; and as an independent consultant to its principal investor.

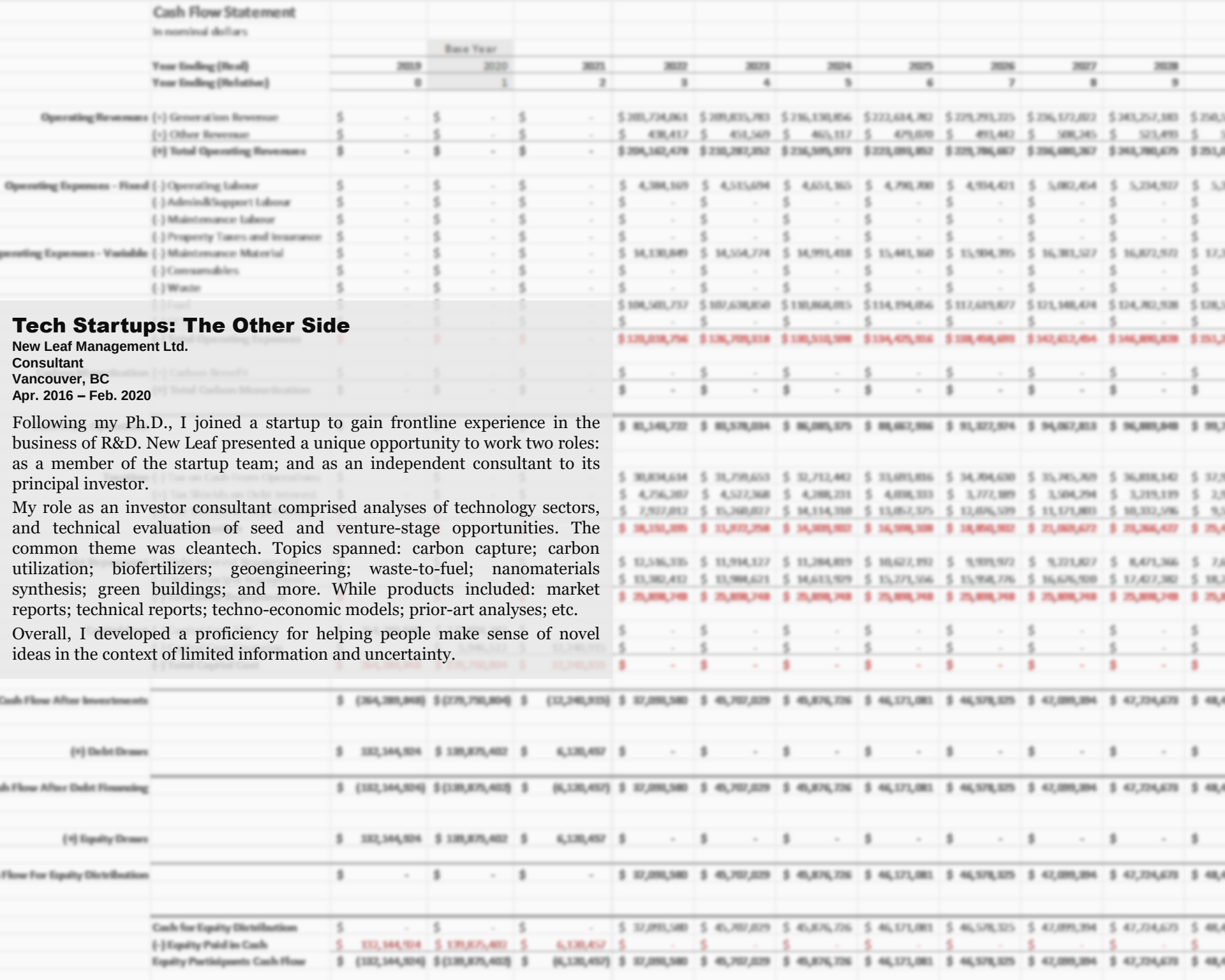
New Leaf pivoted several times as it searched for profitability. Projects included:

- Development of a novel rapid-deployment vertical axis wind turbine. I was hired to support the scale-up to 100's of kW, where I was responsible for mechanical design of the turbine's deployment system. Ultimately, the project failed to justify the risk-capital for scale-up.
- Consulting for a thermal power plant retrofit developer, where I was responsible for plant thermodynamic modelling, costing, and financial forecasting. Analyses for several distinct concepts were delivered.
- Developing an airborne wind energy system (pumping kite generator), where my responsibilities included: techno-economic modelling; IP management; supplier and client contract negotiation; and project documentation. Unfortunately, our key client ran out of cash.

I took on progressively greater responsibility over time. My contributions began with mechanical design, and grew by necessity to include business development and strategic planning. Two patent applications were authored, at least one was published.







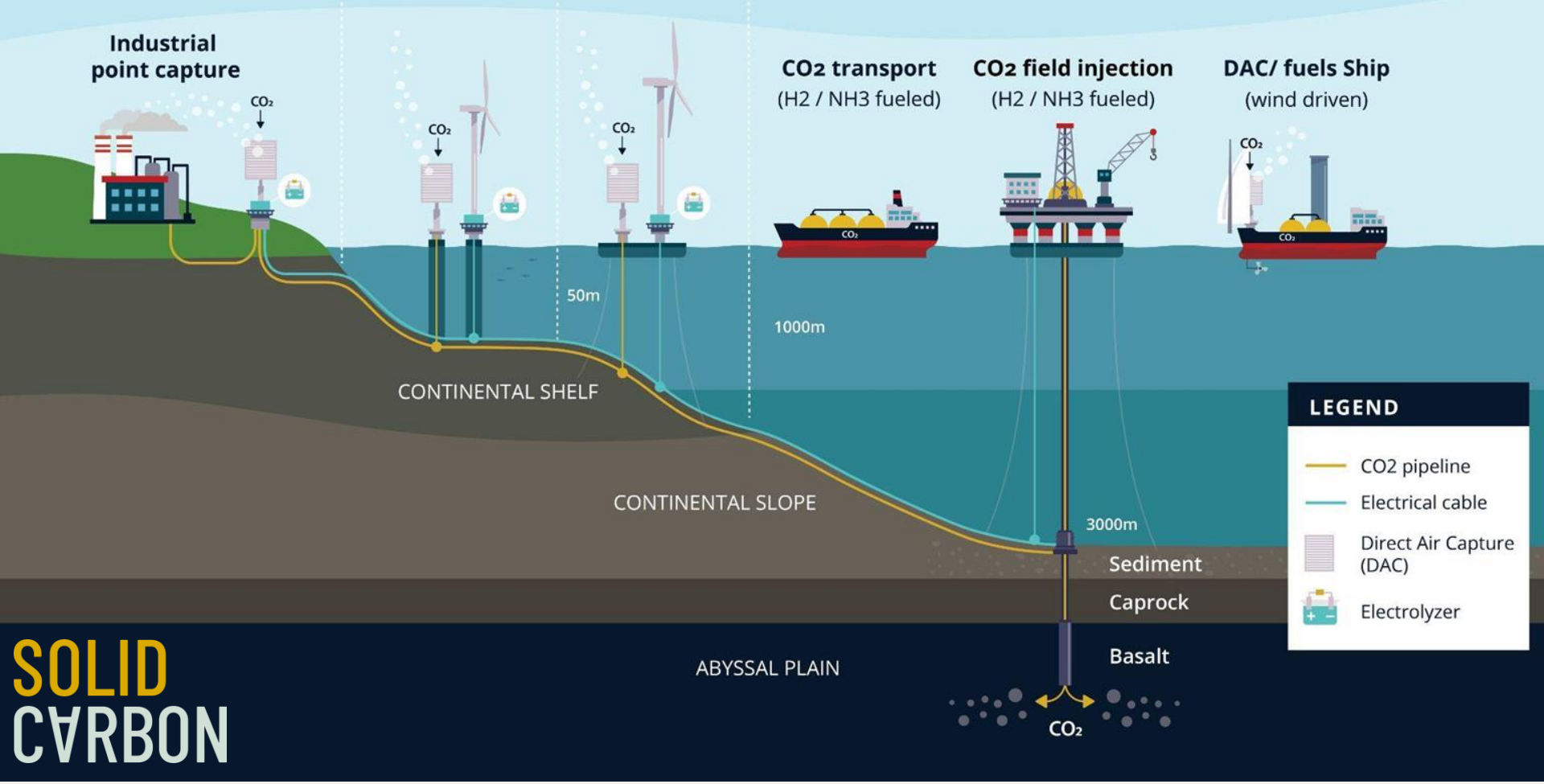
Tech Startups: The Other Side

New Leaf Management Ltd.
Consultant
Vancouver, BC
Apr. 2016 – Feb. 2020

Following my Ph.D., I joined a startup to gain frontline experience in the business of R&D. New Leaf presented a unique opportunity to work two roles: as a member of the startup team; and as an independent consultant to its principal investor.

My role as an investor consultant comprised analyses of technology sectors, and technical evaluation of seed and venture-stage opportunities. The common theme was cleantech. Topics spanned: carbon capture; carbon utilization; biofertilizers; geoengineering; waste-to-fuel; nanomaterials synthesis; green buildings; and more. While products included: market reports; technical reports; techno-economic models; prior-art analyses; etc.

Overall, I developed a proficiency for helping people make sense of novel ideas in the context of limited information and uncertainty.



Putting Carbon Back

Solid Carbon and Pacific Institute for Climate Solutions
Researcher-in-Residence
Victoria, BC
Mar. 2020 – Apr. 2024

The objective was to advance Solid Carbon through a feasibility study. Solid Carbon helps mitigate climate change by removing CO2 from the atmosphere and storing it beneath the seafloor. A defining feature is that stored CO2 reacts with resident basalt minerals and transforms into solid rock.

The Solid Carbon team is an international research partnership, with a core of a couple dozen academic and industry collaborators. The team's work spans three activity areas: systems engineering (1); geoscience and demonstration design (2); and public, regulatory, and investor acceptance (3).

I was brought onboard by Solid Carbon to advance the work funded by PICS' flagship $1 million grant (a large sum for early-stage research). They knew they needed help, but they did not know where or how. I determined the best way to meet the needs of the project was a combination of alliance management and technical work across Solid Carbon's activity areas. My contributions included: developing the systems model (esp. the economics); scoping business development research; developing content for and facilitating public workshops; bridging knowledge and coordinating effort across activity areas; coaching students; and more.

Outcomes of the feasibility study support proceeding to a field demonstration in the Cascadia Basin offshore Vancouver Island. Meanwhile, Solid Carbon has been showcased on TV, radio, newspapers, conferences, etc.. Its future is also a topic of discussion within federal and provincial cabinets.



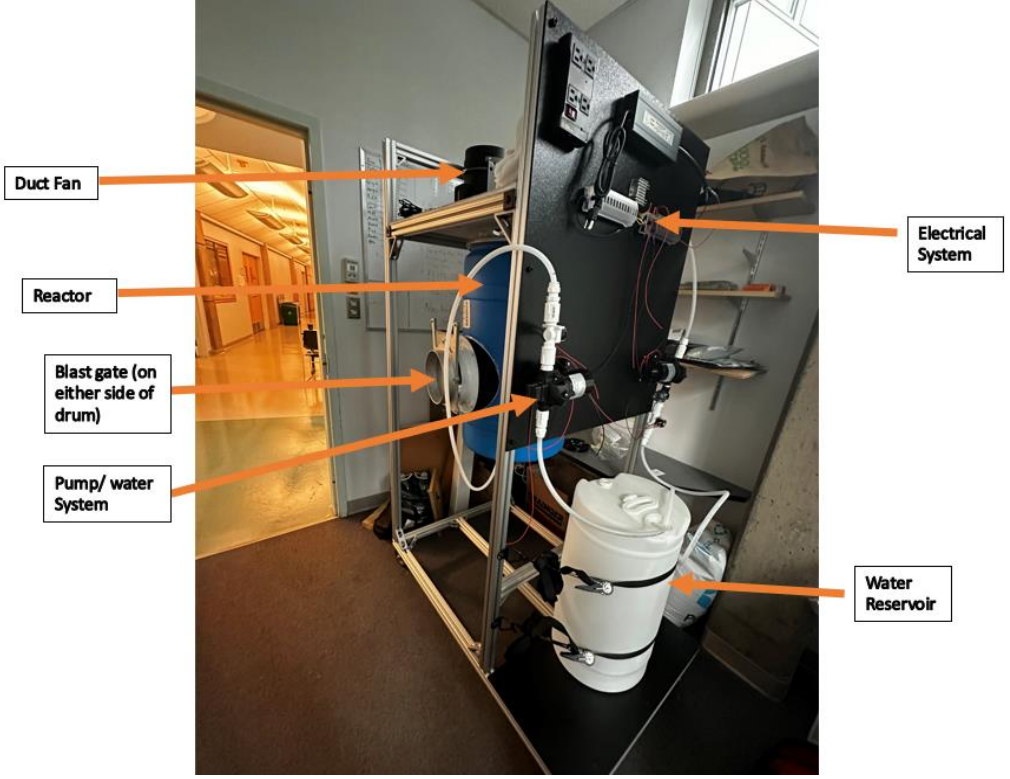
Student Team Mentoring

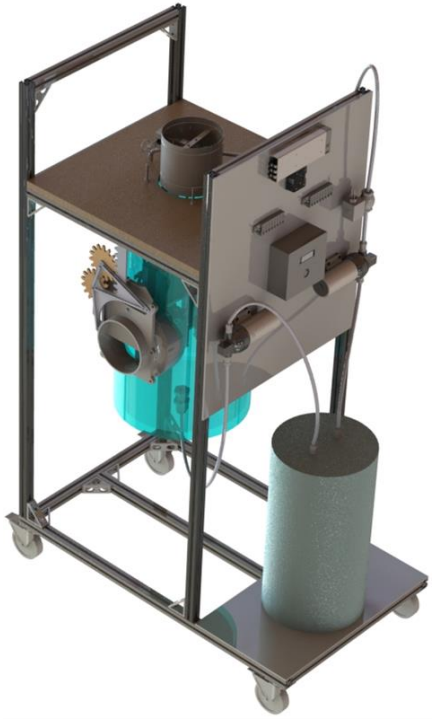
Pacific Institute for Climate Solutions
Researcher-in-Residence
Victoria, BC
Mar. 2021 – Apr. 2023

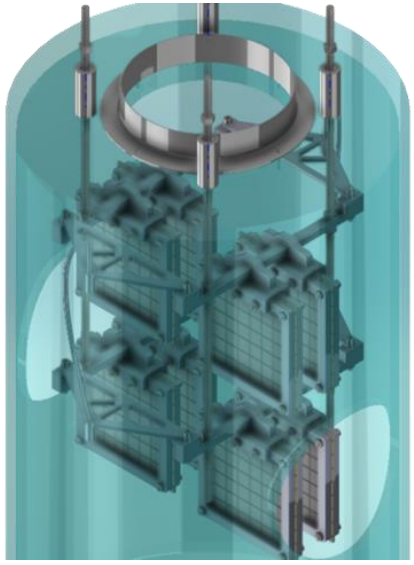
The objective was to provide engineering students experiential learning connected to climate action. A motivated group within the UVic Renewable Energy Club (UREC) sought to do something about CO_2, and approached me and my colleagues at ISEVic. I mentored the students as they set about developing a direct-air capture (DAC) device.

The process began with my helping the students establish their goals and expectations. They identified developing applied skills and building community (esp. given COVID-19's disruptions). I connected the students with the OpenAir Collective, an international network similarly interested in DAC hardware and policy. With advice from the network and myself, the students pursued construction of a moisture swing system which absorbs CO_2 when dry and releases it when wet – an accessible entry point into DAC technology.

The team made great progress towards developing the sorbent cartridge and the balance-of-plant. They were recognized as finalists in the inaugural 2023 Carbon Removal Challenge student competition, and were invited to present at New York University's Washington Square campus. The team has also showcased their work to various leaders, including ministers Heyman (ENV) and Osborne (EMLI). The project remains homed with UREC, with the future to be determined by new student cohorts.







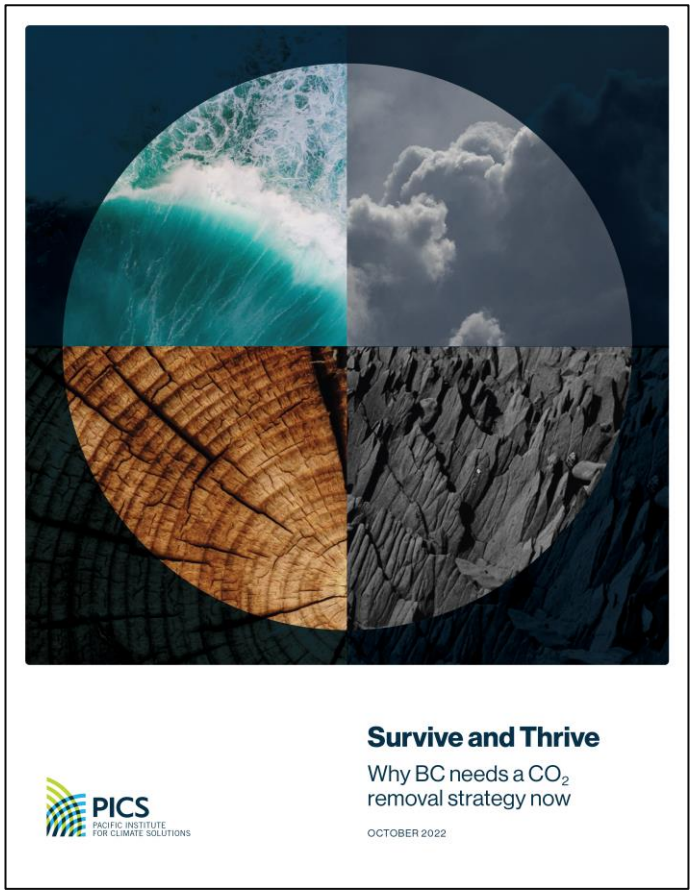
Framing Carbon Management in BC

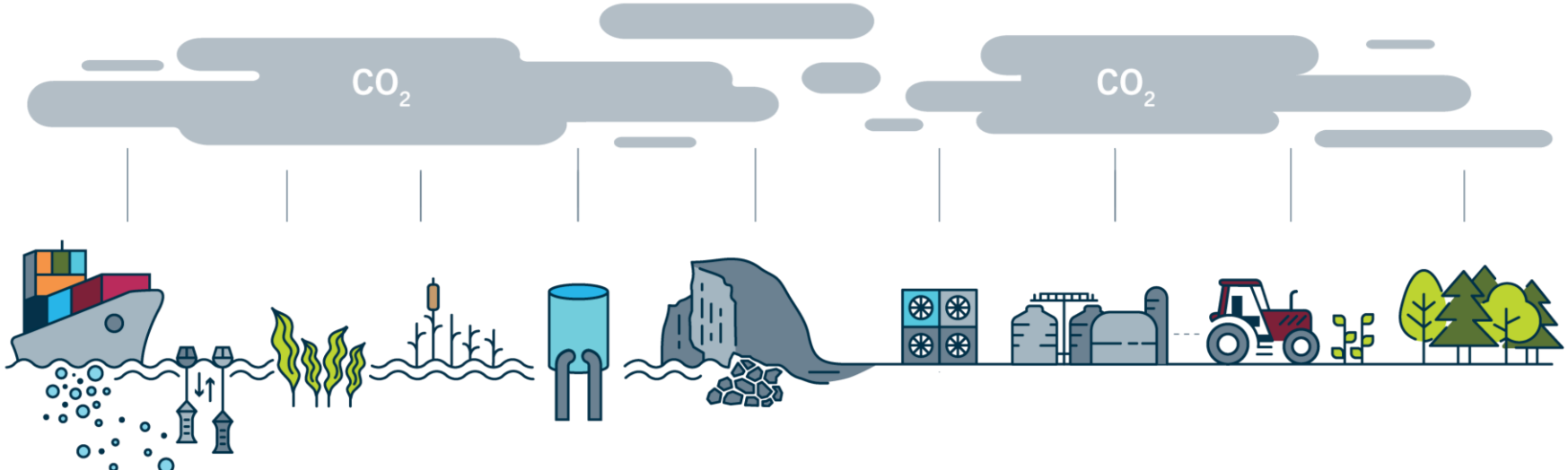
Pacific Institute for Climate Solutions
Researcher-in-Residence
Victoria, BC
Oct. 2022

The objective was to advance discussion of atmospheric carbon dioxide removal (CDR, aka negative emissions technologies, NETs) within BC. Despite being a certainty for any Paris Agreement compatible future, BC had been neglecting the need – which is understandable as the scale, complexity, and implications of CDR are sobering.

The need for a made-in-BC strategy was apparent, so I authored a primer to kickstart its creation. With a focus on getting impact, I crafted a narrative which: tore down disciplinary/technological silos; and built bridges across policy, industry, and research communities. The first half is a call-to-action, it establishes: what is CDR; why it is needed; and how our current trajectory is insufficient. The second half responds to this call-to-action with a strategy framework to build upon.

As a trusted expert in the climate action community, the report succeeded in opening up conversation during its development and post-publication. Uptake included: public webinars; CBC radio interviews; local papers; a Globe and Mail op-ed; and several briefings to BC deputy ministers and assistant deputy ministers. My work's influence is also evident in the BC Centre for Innovation and Clean Energy's subsequent BC Carbon Management Blueprint. Together, we succeeded in shifting policy discussion from a narrow focus on industrial emitters, to one which addresses carbon management comprehensively.





Climate Insights: Climate Education for BC

Pacific Institute for Climate Solutions
Researcher-in-Residence
Victoria, BC
Mar. 2023 – Jan. 2024

The BC Ministry of Forests required climate action training for its workforce. Co-led alongside PICS experts in adaptation and education, I created a four-part eLearning course addressing: climate science, adaptation, mitigation, and taking action.

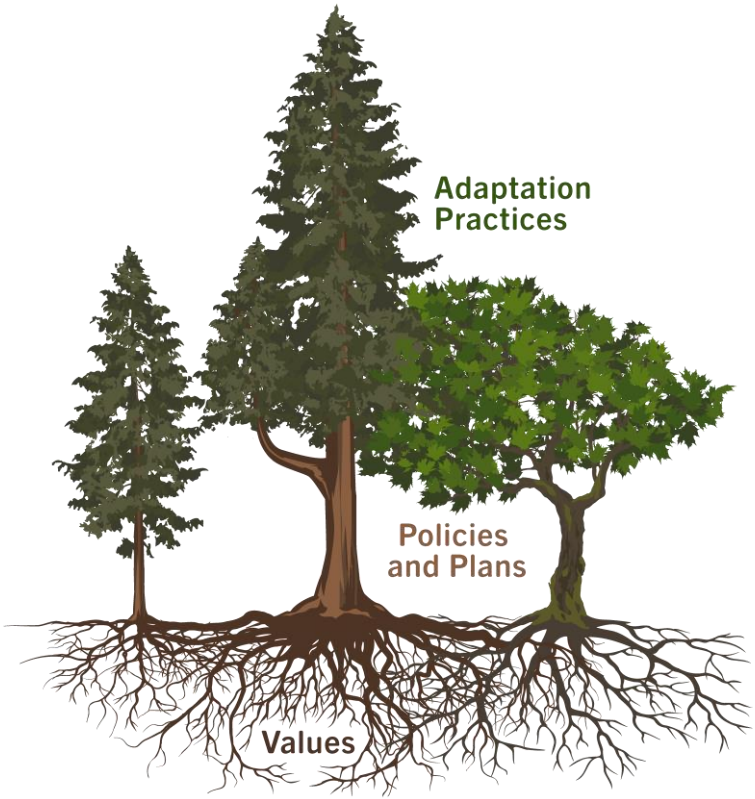
Very much not "just another climate course", we had the ministry's trust to develop a course at the forefront of climate education which:

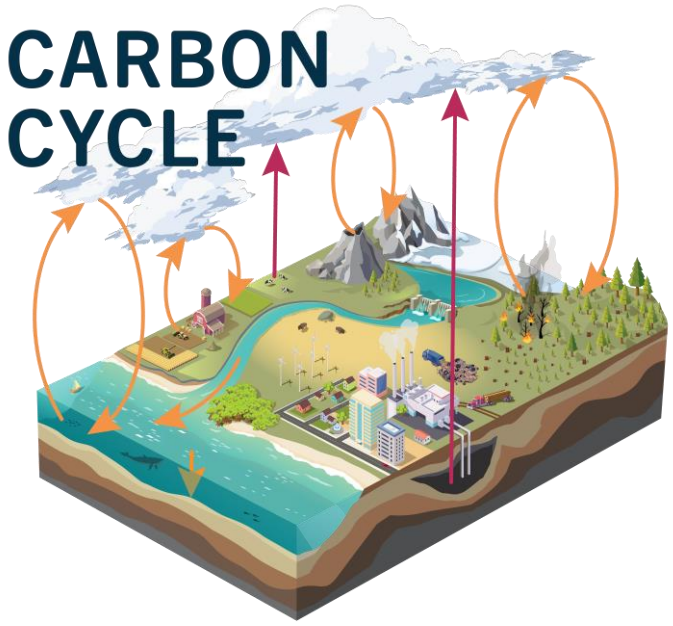
- Embraces the complexity and interconnectedness within climate action and across our broader socioecological system.
- Challenges learners progressively to unlearn siloed-thinking and engage with systemic action.
- Prioritizes not what, but how to think – in a way that can endure an uncertain and evolving world.

My contributions ran from fielding the ministry's first inquiry, through to final release. Our transdisciplinary team shared design and management responsibilities, wherein I also substantively authored the climate science and mitigation modules.

The course forms the foundation for climate action training within Forests, and is being advanced as a requirement for the entire BC public service. It leads the educational pillar of PICS' newest strategic plan, and follow-up products are being discussed with other ministries and NGOs. The course will be available to the public in 2024Q1.







Journal Publications

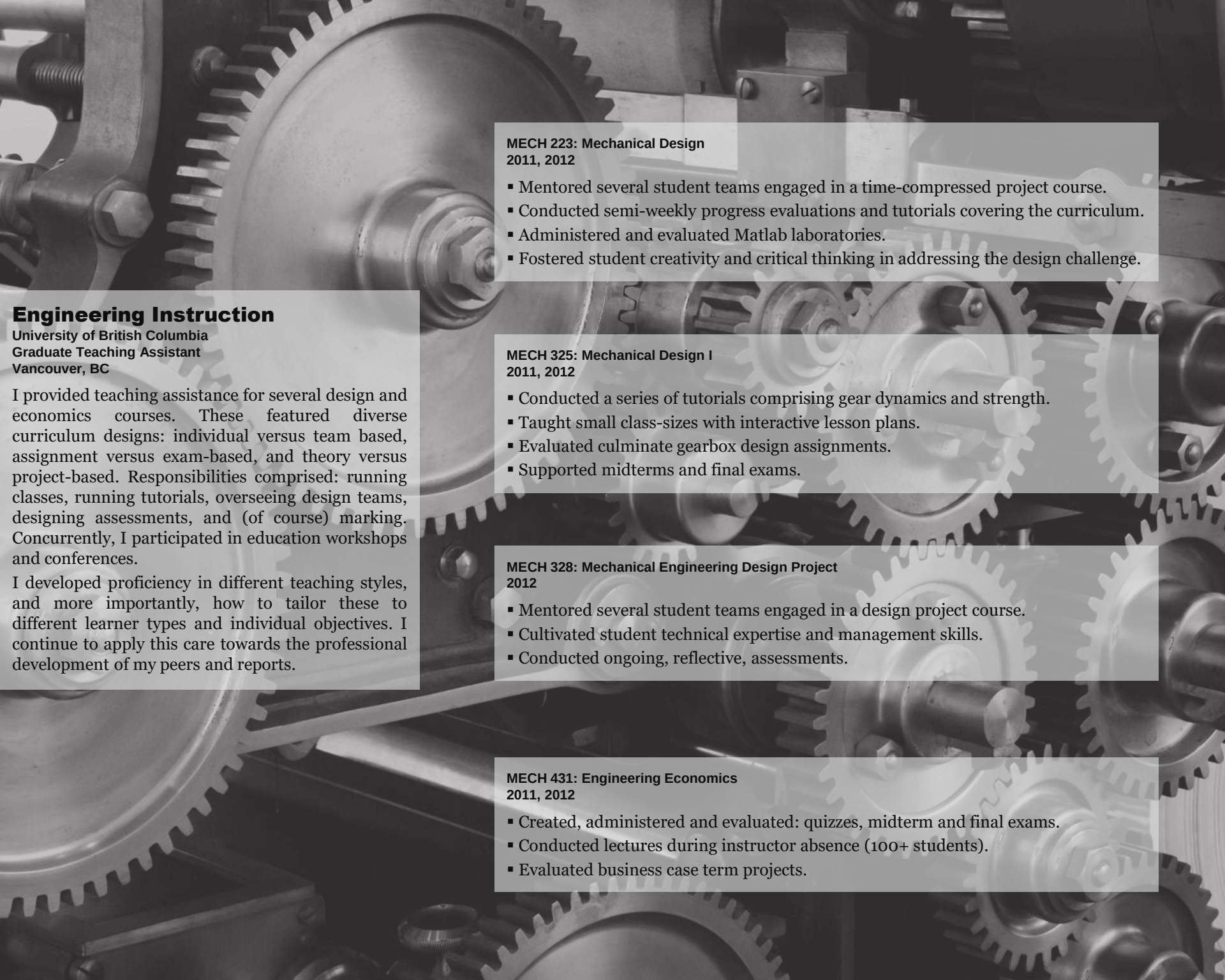
- H. Norton, D. Todd, C. Crawford (2024) “Storage capacity estimates and site conditions of potential locations for offshore-wind powered carbon dioxide removal and carbon sequestration in ocean basalt”. *Carbon Capture Science & Technology*. DOI: 10.1016/j.ccst.2024.100231
- H.M. Teicher, C.A. Phillips, D. Todd (2021) “Climate solutions to meet the suburban surge: leveraging COVID-19 recovery to enhance suburban climate governance”. *Climate Policy*. DOI: 10.1080/14693062.2021.1949259
- D. Todd, W. Mérida. (2016) “Synthesis of transport layers with controlled anisotropy and application thereof to study proton exchange membrane fuel cell performance”. *Journal of Power Sources*. DOI: 10.1016/j.jpowsour.2016.02.031
- D. Todd, S. Bennett, W. Mérida. (2016) “Anisotropic electrical resistance of proton exchange membrane fuel cell transport layers as a function of cyclic strain”. *International Journal of Hydrogen Energy*. DOI: 10.1016/j.ijhydene.2016.02.111
- D. Todd, M. Schwager, W. Mérida. (2015) “Three-dimensional anisotropic electrical resistivity of PEM fuel cell transport layers as functions of compressive strain”. *Journal of The Electrochemical Society*. DOI: 10.1149/2.0611503jes
- D. Todd, W. Mérida. (2015) “Morphologically controlled fuel cell transport layers enabled via electrospun carbon nonwovens”. *Journal of Power Sources*. DOI: 10.1016/j.jpowsour.2014.09.095
- S.R. Dhanushkodi, M. Schwager, D. Todd, W. Mérida. (2014) “PEMFC durability: spatially resolved Pt dissolution in a single cell”. *Journal of The Electrochemical Society*. DOI: 10.1149/2.1031412jes
- D. Todd, M. Schwager, W. Mérida. (2014) “Thermodynamics of high-temperature, high-pressure water electrolysis”. *Journal of Power Sources*. DOI: 10.1016/j.jpowsour.2014.06.144

Other Works

- D. Todd, et. al. (2024) “Climate Insights”. A four-part eLearning course for the BC public service and the general public. https://pics.uvic.ca/resources/courses/
- D. Todd. (Nov 2022) “Negative emissions technologies are crucial if we are to limit global warming”. Op-ed in The Globe and Mail. https://www.theglobeandmail.com/opinion/article-negative-emissions-technologies-are-crucial-if-we-are-to-limit-global/
- D. Todd. (Oct 2022) “Survive and Thrive Why BC needs a CO 2 removal strategy now”. A report by the Pacific Institute for Climate Solutions. https://pics.uvic.ca/sites/default/files/PICS%20Report_Survive%20and%20Thrive%20Oct%202022.pdf
- D. Todd. (Mar 20, 2023) “The IPCC’s newest report and carbon management in BC”. Live interview on CBC’s On the Island.
- D. Todd. (Nov 10, 2022) “On COP27 and mitigating emissions in BC”. Live interview on CBC’s On the Island.
- D. Todd. (Feb 17, 2021) “Episode 9: Sixty ways to hit “net zero” & Negative emissions technologies”. Interview on the Smart Prosperity Institute podcast. https://institute.smartprosperity.ca/Podcast09
- S. Nawaz, D. Todd, S. Satterfield. (Feb 02, 2023) “Reviewer 2 does geoengineering: Breaking! DAC & public opinion”. Interview on the Reviewer 2 podcast. https://podcasters.spotify.com/pod/show/reviewer2geoengineering/episodes/Breaking--DAC--public-opinion---Nawaz--Todd--Satterfield-e1ucd9r
- Columbia World Projects (Oct 2020) “Offshore Carbon Capture and Storage Opportunities and Challenges for CO2 Removal”. Expert participant in Columbia’s workshop. https://worldprojects.columbia.edu/sites/default/files/2021-05/Accelerating%20Offshore%20Carbon%20Capture%20and%20Storage%20Report.pdf
- F. Bourgault, D. Todd, J. Beatch, M. Kheiri, L. Damron, V. Saberi-Nasrabad. (2017) “Method and system for harnessing wind energy using a tethered airfoil”, WO2018213913A1, filed July 17.
- W. Mérida, D. Todd, S. Bennett. (2015) “Electrical resistance characterization of porous conductive media layers”, US provisional patent application 62/189,363, filed July 7.

Talks and Conferences

- B. Tutolo*, M. Scherwath, K. Moran, D. Goldberg, A. Awolayo, L. Coogan, C. Crawford, S. Dosso, E. Ekpo Johnson, R. Lauer, E. Louis, S. Nawaz, T. Satterfield, A. Slagle, D. Todd, R. Webb. (Apr 2023) “Solid Carbon: Safe and Durable Carbon Storage in Ocean Basalt - From Feasibility to Demonstration to Global Potential”. Poster presentation at EGU General Assembly 2023, Vienna, AT.
- R. Webb*, M. Scherwath, K. Moran, D. Goldberg, C. Crawford, B. Tutolo, T. Satterfield, J. Stigant, D. Todd, and the Solid Carbon Team. (Apr 2023) “Solid Carbon: Offshore Negative Emission Technology for Durable Carbon Storage in the Ocean Crust”. Presentation at Ocean Visions 2023.
- M. Scherwath*, K. Moran, D. Goldberg, D. Awolayo, L. Coogan, C. Crawford, S. Dosso, E. Johnson, R. Lauer, E. Louis, T. Satterfield, A. Slagle, D. Todd, B. Tutolo, R. Webb. (Dec 2022) “Solid Carbon: Safe and Durable Carbon Storage in Ocean Basalt - From Feasibility to Demonstration to Global Potential”. Poster presentation at AGU Fall Meeting 2022, Chicago, IL.
- H. Norton*, G. Avellaneda Domene, R. Foxall, P. Gillessen, D. Todd, C. Crawford. (Sep 2022) “Comparative technoeconomic analysis of a floating offshore wind powered direct-air CO2 capture system”. Poster presentation at UMERC+METS 2022, Portland, OR.
- M. Scherwath*, B. Tutolo, D. Goldberg, E. Johnson, K. Khandelwal, R. Lauer, E. Louis, K. Moran, A. Slagle, E. Solomon, D. Todd. (May 2022) “Solid Carbon: A deep-sea CO2 mineralization demonstration project facilitated by IODP data and samples”. Presentation at GAC-MAC-IAH-CNC-CSPG Joint Meeting, Halifax, NS.
- D. Todd*. (2021) “Pathways to net-zero”. Invited talk at the Protect Our Winters (POW) Leadership Summit, Squamish, BC.
- H. Norton*, D. Todd, C. Crawford. (May 2021) “A global assessment of potential locations for offshore wind-powered direct air capture and storage”. Presentation at Wind Energy Science Conference (WESC).
- D. Todd*, C. Crawford, D. Goldberg, K. Moran, T. Satterfield, M. Scherwath, B. Tutolo, R. Webb, and the Solid Carbon Team. (May 2021) “Solid Carbon: A Negative Emissions Technology for CO2 Removal from Air and Durable Storage in Sub-Seafloor Basalts”. Poster presentation at Ocean Visions 2021 Summit.
- D. Todd*. (Apr 2021) “Putting the DAC in CCS: Pieces of the NETs Puzzle”. Presentation at EnVision 2021 Conference, Victoria, BC.
- D. Todd*. (Apr 2021) “PICS, Solid Carbon, and NETs”. Invited talk by EGBC’s climate action working group.
- F. Bourgault*, D. Todd, M. Kheiri, J. Beatch, L. Damron, V. Saberi-Nasrabad. (Oct 2017). “Efficient and power smoothing drive-train concept for pumping kite generators using hydraulics”. Presentation at Airborne Wind Energy Conference, Freiburg, DE.
- D. Todd*, W. Mérida. (Aug 2015) “Synthesis and performance of anisotropic porous transport layers”. Presentation at International Conference on Electrochemical Energy Science and Technology, Vancouver, BC.
- D. Todd*, M. Schwager, L. Damron, S.R. Dhanushkodi, W. Mérida. (Jun 2015) “Sustainable transportation: current developments for the PEM fuel cell at UBC”. Poster presentation at the 7th International Conference on Engineering Education for Sustainable Development, Vancouver, BC.
- D. Todd*, M. Schwager, S.R. Dhanushkodi, W. Mérida. (May 2015) “Porous transport layer characterization and efforts in engineered structures”. Poster presentation at Catalysis Research for Polymer Electrolyte Fuel Cells Network, Burnaby, BC.
- D. Todd*, S.R. Dhanushkodi, W. Mérida. (May 2015) “Progress in PEM fuel cell diagnostics and characterization”. Presentation at Catalysis Research for Polymer Electrolyte Fuel Cells Network, Burnaby, BC.
- D. Todd*, M. Schwager, S.R. Dhanushkodi, W. Mérida. (Mar 2015) “Recent development for the PEM fuel cell at UBC”. Poster presentation at the 1st UBC Sustainability Research Symposium, Vancouver, BC.
- L. Damron*, M. Schwager, S. Flick, D. Todd, S.R. Dhanushkodi, W. Mérida. (Oct 2014) “The recent development of diagnostic tools for the PEM Fuel cell catalyst layer at UBC”. Poster presentation at International Symposium on Electrocatalysis, Whistler, BC.
- D. Todd*, S. Flick, W. Mérida. (Jun 2014) “UBC activities within the CARPE-GECKO collaboration”. Presentation at Catalysis Research for Polymer Electrolyte Fuel Cells Network, Burnaby, BC.
- M. Schwager*, D. Todd, S.R. Dhanushkodi, W. Mérida. (Jun 2014) “Spatially resolved PEMFC diagnostics via segmented cell featuring independent power electronics applied to water and catalyst management”. Presentation at the 97th Canadian Chemistry Conference and Exhibition, Vancouver, BC.
- S.R. Dhanushkodi*, M. Schwager, D. Todd, W. Mérida. (Jun 2014) “Use of segmented cells in durability testing: experimental studies to measure the spatial distribution of Pt surface area and performance loss in PEM fuel cells”. Presentation at the 97th Canadian Chemistry Conference and Exhibition, Vancouver, BC.
- S. Bennett*, D. Todd, W. Mérida. (Mar 2014) “Analysis of the effect of plastic deformation on the through plane conductivity of porous transport layers in proton exchange membrane fuel cells”. Presentation at UBC Multidisciplinary Undergraduate Research Conference, Vancouver, BC.
- O.E. Herrera, D. Todd*, S. Flick, W. Mérida. (May 2011) “Heat management in anode and cathode of a PEMFC”. Presentation at Hydrogen + Fuel Cells 2011: International Conference and Exhibition, Vancouver, BC.



Engineering Instruction

University of British Columbia
Graduate Teaching Assistant
Vancouver, BC

I provided teaching assistance for several design and economics courses. These featured diverse curriculum designs: individual versus team based, assignment versus exam-based, and theory versus project-based. Responsibilities comprised: running classes, running tutorials, overseeing design teams, designing assessments, and (of course) marking. Concurrently, I participated in education workshops and conferences.

I developed proficiency in different teaching styles, and more importantly, how to tailor these to different learner types and individual objectives. I continue to apply this care towards the professional development of my peers and reports.

MECH 223: Mechanical Design
2011, 2012

- Mentored several student teams engaged in a time-compressed project course.
- Conducted semi-weekly progress evaluations and tutorials covering the curriculum.
- Administered and evaluated Matlab laboratories.
- Fostered student creativity and critical thinking in addressing the design challenge.

MECH 325: Mechanical Design I
2011, 2012

- Conducted a series of tutorials comprising gear dynamics and strength.
- Taught small class-sizes with interactive lesson plans.
- Evaluated culminate gearbox design assignments.
- Supported midterms and final exams.

MECH 328: Mechanical Engineering Design Project
2012

- Mentored several student teams engaged in a design project course.
- Cultivated student technical expertise and management skills.
- Conducted ongoing, reflective, assessments.

MECH 431: Engineering Economics
2011, 2012

- Created, administered and evaluated: quizzes, midterm and final exams.
- Conducted lectures during instructor absence (100+ students).
- Evaluated business case term projects.

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- The New Leaf photograph is copyright of New Leaf. It depicts a VAWT I was to scale-up.
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- The UREC illustrations were created by student team members.
- The NET report and climate course illustrations are copyright of PICS. I worked with an illustrator to create them.
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